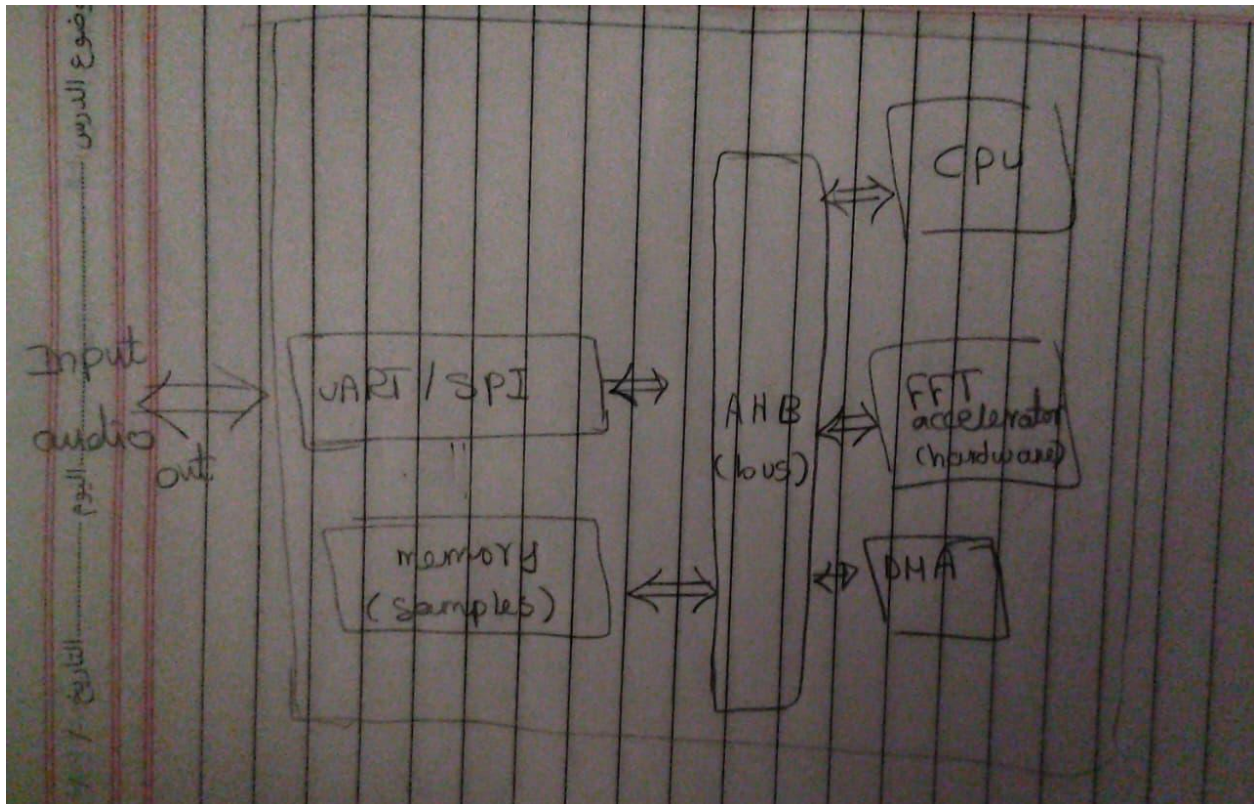


SOC lab

Name: Hagar Emadeldin Emara



1. Store samples in memory --- HW: DMA writes incoming UART/SPI bytes into data memory automatically (speed)
2. Apply optional windowing --- SW, runs on the CPU core, reading/writing the sample buffer in data memory (repetitive)
3. Compute FFT---- HW, offloaded to the dedicated accelerator (high computation)
4. Compute magnitude spectrum --- SW, CPU reads the FFT accelerator's output registers/buffer and computes. (simple computation)
5. Find peak bin --- SW, CPU scans the magnitude array in memory (simple computation)
6. Convert peak bin to frequency --- SW, a single multiply on the CPU (simple computation)

Algorithm:

Wait for new block of samples to arrive

Store samples in memory

IF windowing is enabled:

Smooth edges of the sample block

Send samples to FFT hardware

Wait until FFT is done

Get frequency components from FFT

For each frequency component:

 Compute its magnitude (strength)

Find the component with the highest magnitude

Convert its position into a real frequency value

Send frequency value back to host